Pre-clinical Operative Dentistry

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Composite resin filling materials

Indications for composite:

1. Classes I, II, III, IV, V and VI restorations
2. Core buildups
3. Fissure sealants and conservative composite restorations (preventive resin restorations)
4. Esthetic enhancement procedures
   - Partial veneers
   - Full veneers
   - Tooth contour modifications
   - Diastema closures
5. Cements (for indirect restorations)
6. Temporary restorations

Contraindications:

1. An operating area that cannot be adequately isolated.
2. Class V restorations that are not aesthetically critical.
3. Restorations that extend into the root surface (may exhibit gap formation).
4. When heavy occlusal stresses are present.

Advantages:

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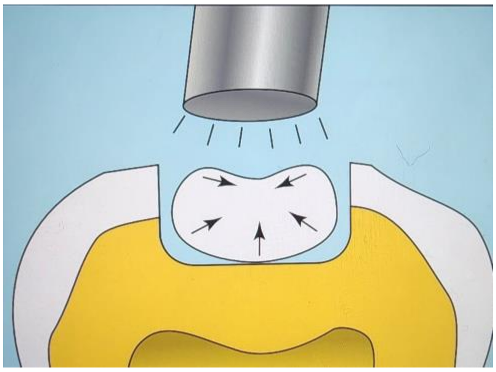
1. Aesthetics
2. Conservative of tooth structure removal (less extension; uniform depth not necessary; mechanical retention usually not necessary)
3. Easier, Less complex when preparing the tooth
4. Low thermal conductivity
5. Bonded to tooth structure resulting in good retention, low micro leakage, minimal interfacial staining, and increased strength of remaining tooth structure
6. Repairable

Disadvantages

- ❖ Material related disadvantages:
    1. Polymerization shrinkage.



    2. Occlusal wear in areas of high occlusal stress.
    3. Have a higher linear coefficient of thermal expansion.
- ❖ Technique related disadvantages:
    1. Time-consuming.
    2. Technique sensitive.
    3. Costly (compared to amalgam restorations)

Composition

1. **Organic Resin**: e.g. dimethacrylate monomer (BIS-GMA)

Function:

The monomer system represents as the backbone of the composite resin system.

2. **Inorganic filler**: e.g. fused silica, crystalline quartz, lithium aluminum silicate, borosilicate glass

Function:

- Reduces polymerization shrinkage.
- Reduce the coefficient of thermal expansion of the resin matrix
- Better mechanical properties, such as compressive strength.
- Greater aesthetics.
- Provides radio-opacity.
- Improves handling properties.

3. **Coupling Agent:** silane coupling agents

- Bond the resin with the filler
- Transfer forces from flexible resin matrix to stiffer filler particles.
- Prevent penetration of water along filler resin interface.

4. **Initiator System** – activate the setting mechanism
5. **Stabilizers**
6. **Pigments**

Classification

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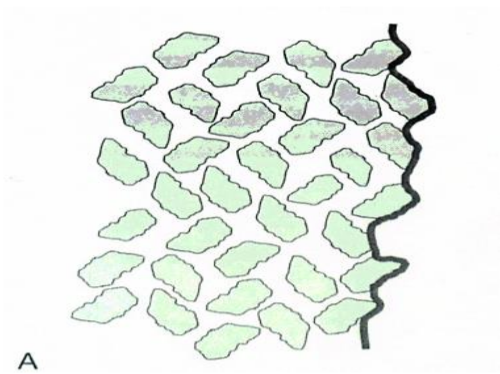
Composite resins have been classified in different ways:

1. According to filler size:
   - Macrofilled
   - Microfilled
   - Hybrid
   - Nano-composite (nanofilled).
2. Classification according to polymerization method:
   - Light cure.
   - Chemical cure.
   - Dual cure.
3. Classification according to matrix compositions
   - BIS-GMA.
   - UDMA.

Conventional Composites

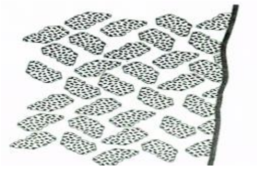
1. contains 75-80% inorganic filler by weight
2. average particle size 8µm
3. large size particle and extremely hard filler
4. rough surface structure, strontium and barium glass (radiopaque)



Microfilled Composites

1. Introduced in 1980

2. Highly polishable, the small particle size results in smooth polished surface which is resistant to plaque, debris and stains
3. Good aesthetic.
4. particle size is 0.01 – 0.04µm
5. Filler content of microfilled resins is 35–50% by weight.
6. Because of less filler content, some of their physical properties are inferior.

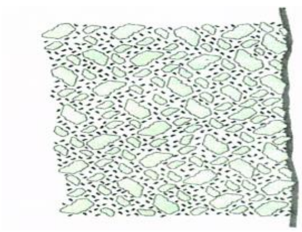


Use of Microfilled Composites

- used for low stress restorations, buccal and lingual surfaces of class III and class V

Hybrid Composites

1. combines the properties of conventional and microfilled
2. contains 75-85% inorganic filler by weight
3. particle size is 0.4 – 1µm
4. physical properties is superior to conventional
5. predominant direct aesthetic resin
6. have universal clinical applicability



Use of Hybrid Composites

- Used in moderate stress restorations where strength and wear resistance are more important than surface luster; Class I, class II, class IV.

Nanofill composites

1. Contain filler particles that are extremely small (0.005-001 microm.)
2. Because of these small particles a high filler levels can be generated in the restorative material, resulting in good physical properties and esthetics
3. Nanofills highly polishable
4. These materials are likely to become a popular composite restorative material choice.

Flowable composites

1. Flows into cavity due to lower viscosity
2. Have lower filler content
3. Inferior physical properties (lower wear resistance, lower strength)
4. Used in small class I, pit and fissure sealant, marginal repair materials, as the first increment placed as a liner under hybrid or packable composites
5. Easy to use
6. Good wet ability
7. Favourable handling properties are popular features

Packable (Condensable) composites

1. More viscous, “thicker, stiffer feel”
2. Have filler particle feature that prevents sliding of the filler particle by one another
3. Easier restoration of proximal contact
4. Similar to the handling of amalgam

Reinforced composites

It consists of a combination of a resin matrix, randomly orientated E-glass fiber and inorganic particulate fillers.

Used as base filling material in high stress bearing areas especially in large cavities of vital and non- vital posterior teeth

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Classification of composite according to the method of activation:

1. Chemically-activated composites:

Also they are called self -curing composite resins. Most commonly available as two-paste system composed of a catalyst and base materials. When these two components are property mixed, the polymerization process is chemically activated. The rate of set is uniform through the bulk of the material causing a gradual increase in viscosity at room temperature. Hence the material have a limited working time, making the technique time sensitive with the increased possibility of air bubble incorporation during mixing of the two pastes and thus affecting the composite physical and mechanical properties .

2. Light-activated: composites:

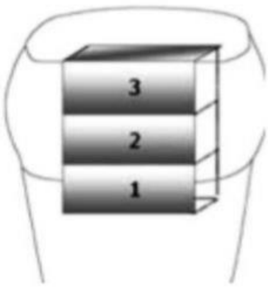
Light activated materials afford a number of advantages over chemically activated ones. The light curable materials are single components, and require no mixing, and so have reduced porosity, and better resistance to wear and abrasion. The working time is virtually that chosen by the clinicians, and the material hardens rapidly when exposed to light. The components of light -activated composites are contained in single paste system. The mixture is supplied in various shades in disposable syringes. These syringes are made of opaque plastic to protect the material from exposure to light.

3. Dual cured composites:

Combine self-curing and light curing materials .The self-curing rate is slow and is designed to cure only those portions that are not adequately light cured specially in the interproximal areas where the access is limited and require special approaches to guarantee adequate light curing energy.

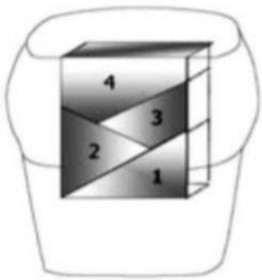
Posterior Composite Placement Techniques
Layering (common techniques)

- ❖ The thickness of each increment of resin composite is not more than 2 mm.
- ❖ Each increment should be fully polymerized before the next one is inserted into the cavity.



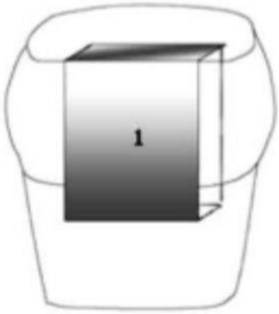
Wedge-Shape Layering (Oblique)

- ❖ In this technique wedge-shaped composite increments are placed and polymerized only from the occlusal surface.



Bulk Technique

- ❖ Its include one bulk increment (4 – 5 mm) without traditional layering,
- ❖ This is done by special composite resin system that's in different ways.



Sandwich technique:

- ❖ In this technique two different materials will be used
- ❖ Glass ionomer or Flowable composite used as liner under composite.
- ❖ Two sandwich technique available:
    1. Closed sandwich technique
    2. Opened sandwich technique
- ❖ Sandwich technique mainly used when the gingival margins extend beyond cement-enamel junction.